

Math 196 Lecture Notes: Week 2

Miles Kretschmer

March 2026

0.1 Review of Last Week

0.1.1 Algorithm to put a matrix in row echelon form

What follows is a more detailed description of the process of putting a matrix into reduced row echelon form by applying row operations.

By the *leading nonzero element* of a row we mean the first nonzero element from left to right. If this is 1 we call it a leading 1. A column is a pivot column if it contains a leading 1 and has zeros in all other places. Notice that this depends only on the column in question and columns to the left of it. In the algorithm, we go through the columns from left to right.

1. If the column is already a pivot column, we are done, and we move on.
2. If the column contains no leading nonzero element, we cannot make it into a pivot column, and we move on.
3. If the column contains a leading nonzero element a of a row, we make the column a pivot in two steps:
 - (a) Multiply the row by $1/a$ to make the leading element a 1.
 - (b) Subtract multiples of the row with the leading 1 from other rows to remove all other elements of the column.

Once the column is a pivot, we reorder the rows to move the 1 up below the previously created leading 1s.

0.1.2 Interpreting the row echelon form: the meaning of pivot variables and free variables

Once the coefficient matrix is in reduced row echelon form, we can translate everything back into a system of equations.

At the bottom, there may be equations $0 = a$ for some a , corresponding to 0 rows in the coefficient matrix. This means that the system may not be consistent. If any such equation sets 0 equal to something nonzero, the system is inconsistent. Otherwise it is consistent.

We can distinguish between two types of variables

1. Variables corresponding to pivot columns
2. Variables corresponding to non pivot columns

The presence of non-pivot variables means that we do not have a unique solution: if the system is consistent, we get a solution by choosing arbitrary values for the non pivot variables, which then determine values of the pivot variables. Previously, I wrote this schematically by setting the non-pivot variables equal to new letters r, s, t . This gives a concise description of *all* the solutions to the equation.

Compare the case of a single variable equation: if $0 \cdot x = 0$, then x is a non-pivot variable, it can be set arbitrarily.

0.1.3 Review of matrix-vector multiplication

Given a matrix ($m \times n$, m rows and n columns)

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

we can view its columns as vectors, and write $A = \begin{bmatrix} \vdots & \vdots \\ v_1 & v_n \\ \vdots & \vdots \end{bmatrix}$ Here v_j is the vector

$$\begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix} \in \mathbb{R}^m$$

Given a vector

$$w = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

we defined the matrix vector product as a linear combination:

$$Aw = b_1 v_1 + \dots + b_n v_n \in \mathbb{R}^m$$

The i th entry of this is

$$\sum_j a_{ij} c_j$$

Conclusion. Suppose $v_1, \dots, v_n, v$ are vectors with m entries. We let

$$A = \begin{bmatrix} \vdots & \vdots \\ v_1 & \dots & v_n \\ \vdots & \vdots \end{bmatrix}$$

be the matrix with columns $v_1, \dots, v_n$. The following 3 problems are equivalent:

1. Solve the system of linear equations with augmented matrix

$$[A \mid v]$$

2. Write v as a linear combination of $v_1, \dots, v_n$.

3. Find a vector $w \in \mathbb{R}^n$ such that $Aw = v$.

In this formulation, each row of A corresponds to an equation.

1 Matrix-Vector Multiplication and Linear Transformations

1.1 Elementary Properties of Matrix-Vector Multiplication

1.1.1 Linearity Properties

We will need the distributive property for scalar multiplication:

$$(a + b)v = av + bv$$

This follows by applying the distributive property to each entry of the vector.

The matrix vector product satisfies some algebraic properties, like multiplication of numbers:

1. Distributivity: $A(v + w) = Av + Aw$
2. Scalar multiplication: $A(a \cdot v) = a \cdot Av$

Proof of (1). Write

$$A = \begin{bmatrix} \vdots & & \vdots \\ u_1 & \dots & u_n \\ \vdots & & \vdots \end{bmatrix}, \quad v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad w = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Then we have

$$A(v + w) = A \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix} = (x_1 + y_1)u_1 + \dots + (x_n + y_n)u_n$$

Using the distributive property for scalar multiplication of vectors, this is equal to

$$x_1u_1 + y_1u_1 + \dots + x_nu_n + y_nu_n$$

We can rearrange this sum, putting all the x s together and all the y s together, to get

$$(x_1u_1 + \dots + x_nu_n) + (y_1u_1 + \dots + y_nu_n)$$

By definition of the matrix-vector product, this is equal to $Av + Aw$. □

1.1.2 Some Special Products

Recall that last week we introduced special vectors e_i , where e_i has a 1 in the i th position and 0s elsewhere.

Question. What is Ae_i ?

Using the linear combination definition of the matrix vector product, this comes out to exactly u_i , the i th column.

Example 1.

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

We also introduced the *identity matrix* in some examples. In general, I_n is a square matrix

$$I_n = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$$

with n rows and n columns, 1s on the diagonal, and 0s elsewhere (unwritten above). This is exactly the matrix whose i th column is e_i .

Claim. For any vector $v \in \mathbb{R}^n$, $I_nv = v$.

Proof. Let v have entries

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Then

$$I_n v = x_1 e_1 + \cdots + x_n e_n = \begin{bmatrix} x_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \cdots + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ x_n \end{bmatrix} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = v$$

□

1.2 Linear Transformations and Matrices

To a $m \times n$ matrix A we can associate a function T taking a vector $v \in \mathbb{R}^n$ to the vector $T(v) = Av \in \mathbb{R}^m$; we use the notation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ to express this.

Remark. Notice that we are not forgetting anything about the matrix when we view it as a function, the columns of the matrix are exactly $T(e_i)$, so the matrix is still there in the function.

The properties of matrix-vector multiplication we discussed previously can be viewed as properties of this function.

1. $T(v + w) = T(v) + T(w)$
2. $T(a \cdot v) = a \cdot T(v)$

Definition. We call a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ a *linear transformation* if it has these properties:

1. $f(v + w) = f(v) + f(w)$
2. $f(a \cdot v) = a \cdot f(v)$

If you think back to our geometric definitions of vector addition and scalar multiplication, we can see that some geometrically defined transformations on the plane $\mathbb{R}^2$ are linear transformations. For example, rotation about the origin through angle $\pi/2$ (a quarter turn) can be seen to be linear. [Draw Pictures] We will study more examples like this in a moment, but first, we need to clarify the relationship between linear transformations and matrices.

Theorem 1. 1. If A is an $m \times n$ matrix, then the function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(v) = Av$ is a linear transformation.

2. Conversely, if $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear, then we can form a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ T(e_1) & \cdots & T(e_n) \\ \vdots & & \vdots \end{bmatrix}$$

with columns $T(e_i)$. We have $T(v) = Av$ for any v , so in fact T comes from a matrix as in (1).

This tells us that linear transformations and matrices are equivalent; they are two different points of view on the same.

Proof. The only thing left to prove is that if we define A as in (2), then we have $T(v) = Av$ for any v . Write

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

We can view v as a linear combination of the standard basis: $v = x_1 \cdot e_1 + \cdots + x_n \cdot e_n$. Using the fact that T is linear, we have

$$T(v) = T(x_1 \cdot e_1 + \cdots + x_n \cdot e_n) = x_1 \cdot T(e_1) + \cdots + x_n \cdot T(e_n) = Av$$

□

Example 2. The quarter turn linear transformation corresponds to the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

The fact that matrices define linear transformations tells us something about linear systems. Consider a linear system in matrix form: $Ax = b$. If we have a solution v to this, and a solution w to a system $Ax = c$ with the same coefficient matrix, then $v + w$ is a solution to $Ax = b + c$. This is sometimes called the “principle of superposition.” A particular case of interest is when v is a solution to $Ax = b$, and w satisfies $Aw = 0$. Then for any scalar $k \in \mathbb{R}$, $v + k \cdot w$ is a solution to $Ax = b$:

$$A(v + k \cdot w) = Av + A(k \cdot w) = b + k \cdot Aw = b$$

1.3 Geometric Examples of Linear Transformations

Knowing now that linear transformations and matrices are equivalent, we can investigate some geometrically defined linear transformations of $\mathbb{R}^2$, and see how to represent them as matrices.

1.3.1 Scalar Matrices

One of the simplest transformations we can do is to uniformly scale everything by some scale factor a : this is the function $f(v) = a \cdot v$. We can see, both algebraically and geometrically, that this is linear:

$$a \cdot (v + w) = av + vw$$

(look at each component of the vector), and $a \cdot (kv) = ak \cdot v = k \cdot (a \cdot v)$. To represent this as a matrix, we look at what it does to each e_i : these are mapped to ae_i , so we get

$$\begin{bmatrix} a & & \\ & \ddots & \\ & & a \end{bmatrix}$$

We call these “scalar matrices.”

1.3.2 Scaling One Axis

The matrix

$$\begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$$

Represents the transformation

$$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} ax \\ y \end{bmatrix}$$

which scales the x coordinate by a and leaves the y coordinate alone. Likewise, the transformation

$$\begin{bmatrix} 1 & 0 \\ 0 & a \end{bmatrix}$$

scales the y coordinate by a , and leaves the x coordinate alone.

1.3.3 Rotation Matrices

The geometric argument that a quarter turn is a linear transformation applies to all rotations. To find the matrix representation, we can use some trigonometry. [Draw unit circle picture]

$$R_\theta(e_1) = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}$$

$$R_\theta(e_2) = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$

The rotation matrix is

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

Example 3. Find the rotation of

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

through an angle of $\pi/4$. This is the matrix vector-product

$$\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -\sqrt{2} \\ 3\sqrt{2} \end{bmatrix}$$

1.3.4 Some Reflections

We can also find matrix representations for the reflections across the x and y axes. These are respectively

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Later on, we will see how to represent reflection across an arbitrary line through the origin as a reflection.

1.3.5 Shear Matrices

We now consider the matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

We will interpret geometrically what this is doing. It will be instructive to consider how it acts on a square with corners

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Linear transformations map lines to lines, so we can get an idea of what the matrix does to the square by looking at what it does to the corners:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

[Draw Picture]. In general, this transformation acts on an arbitrary point by

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + y \\ y \end{bmatrix}$$

Each point stays at the same vertical level, and shifts horizontally according to its vertical level. We call this a *horizontal shear*. Likewise

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

represents a *vertical shear*. [Draw Picture] In general, matrices

$$\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix}$$

for $a \in \mathbb{R}$ represent respectively horizontal, vertical shears of varying magnitude.

We will see more geometric examples on later on.

2 Composition of Transformations and Matrix Multiplication

2.1 Composition of Linear Transformations

We now return to considering linear transformations in general. Suppose we have a function $f : \mathbb{R}^p \rightarrow \mathbb{R}^q$ and another function $g : \mathbb{R}^q \rightarrow \mathbb{R}^r$. Then, given a vector $v \in \mathbb{R}^p$, we can apply f to get $f(v) \in \mathbb{R}^q$, to which we can apply g to get a vector $g(f(v)) \in \mathbb{R}^r$. This defines a composite function $h = g \circ f : \mathbb{R}^p \rightarrow \mathbb{R}^r$, given by

$$h(v) = g(f(v))$$

If f and g happen to be *linear* functions, h will be as well.

Proposition. *If f and g are linear functions, then their composition is linear.*

Proof. Let $v, w \in \mathbb{R}^p$ and $a \in \mathbb{R}$. Then

$$g \circ f(v + w) = g(f(v + w)) = g(f(v) + f(w)) = g(f(v)) + g(f(w)) = g \circ f(v) + g \circ f(w)$$

$$g \circ f(av) = g(f(av)) = g(af(v)) = ag(f(v)) = a \cdot g \circ f(v)$$

□

2.2 Matrix Multiplication

Because linear transformations $\mathbb{R}^p \rightarrow \mathbb{R}^q$ are equivalent to $q \times p$ matrices, the fact that linear transformations can be composed should correspond to an operation on matrices. In particular, given a $q \times p$ matrix A and a $r \times q$ matrix B , we can form the associated linear transformations $T_A : \mathbb{R}^p \rightarrow \mathbb{R}^q$ and $T_B : \mathbb{R}^q \rightarrow \mathbb{R}^r$, then compose them to get $T_B \circ T_A : \mathbb{R}^p \rightarrow \mathbb{R}^r$. This is linear, and so should correspond to a $r \times p$ matrix C .

How do we find C ? We need to know what is columns are, and we find these by applying $T_B \circ T_A$ to the standard basis vectors $e_i \in \mathbb{R}^p$. $T_B \circ T_A(e_i) = T_B(T_A(e_i))$. We saw that $T_A(e_i)$ is exactly the i th column of A , v_i , so this comes out to the matrix vector product Bv_i . In other words, the obtain the columns of C , we apply B to the columns of A .

Definition. Given a $q \times p$ matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_p \\ \vdots & & \vdots \end{bmatrix}$$

and a $r \times q$ matrix B , we define the product BA to be the matrix

$$\begin{bmatrix} \vdots & & \vdots \\ Bv_1 & \dots & Bv_p \\ \vdots & & \vdots \end{bmatrix}$$

As each $Bv_i \in \mathbb{R}^r$, this is an $r \times p$ matrix, representing a linear function $\mathbb{R}^p \rightarrow \mathbb{R}^r$ which is the composition of the functions represented by A and by B .

Remark. One could unwrap the definition of the matrix vector product to get a formula for each individual entry of BA in terms of entries of B and entries of A . I find the formula in terms of applying B to the columns of A easier to remember and apply.

Example 4. Consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 1 \end{bmatrix}$$

The product AB is not well defined: B outputs vectors in $\mathbb{R}^3$, but A takes as input vectors in $\mathbb{R}^2$. The product BA is well defined: A outputs vectors in $\mathbb{R}^2$, which B takes as input. To compute BA , we apply B to each column of A :

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} &= \begin{bmatrix} 5 \\ -1 \\ 3 \end{bmatrix} \end{aligned}$$

Therefore, the product is

$$BA = \begin{bmatrix} 1 & 5 \\ 1 & -1 \\ 0 & 3 \end{bmatrix}$$

Example 5. We will compute the transformation of $\mathbb{R}^2$ given a horizontal shear with magnitude 1, followed by a rotation through $\pi/2$. The first of these is

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

and the second is

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

The matrix that is applied first goes on the right, as in function composition $g \circ f$ means to apply f first. We thus compute

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The first column of this is

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The second column is

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Therefore, the composite transformation is given by the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$$

2.3 Properties of Matrix Multiplication

We will now discuss some general algebraic properties of matrix multiplication.

First of all, addition and scalar multiplication makes sense for matrices just as it does for vectors, moreover:

$$\begin{aligned}(A + B)C &= AC + BC \\ A(B + C) &= AB + AC \\ (kA)B &= A(kB) = k(AB)\end{aligned}$$

2.3.1 Identity Matrix

Let A be an $m \times n$ matrix, and I_m be the $m \times m$ identity matrix. Then the product $I_m A$ makes sense as an $m \times n$ matrix: A outputs vectors in $\mathbb{R}^m$, and I_m takes these as input. This product is $I_m A = A$. Indeed, if A has columns $v_1, \dots, v_n$, then the columns of $I_m A$ are $I_m v_1 = v_1, \dots, I_m v_n = v_n$.

Similarly, $A I_n = A$: the columns of I_n are $e_1, \dots, e_n$, and $A e_i$, the i th column of $A I_n$, comes out to exactly the i th column of A .

2.3.2 Associativity

If we consider a product of three matrices ABC , we can compute this either as $(AB)C$ or $A(BC)$. Associativity says that we get the same answer either way:

$$(AB)C = A(BC)$$

A similar conclusion holds for products of any number of matrices. The reason for this property is that matrix multiplication represents composition of functions, and this is associative. If $A \rightsquigarrow f$, $B \rightsquigarrow g$, $C \rightsquigarrow h$, then both $(AB)C$ and $A(BC)$ represent the composite function

$$v \mapsto f(g(h(v)))$$

2.3.3 Matrix Multiplication is Not Commutative

On the other hand, the order of matrix multiplication does matter. First of all, if AB is a well defined matrix product, BA may not be: see

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 0 \end{bmatrix}$$

Even if both AB and BA are defined, they may be matrices of different sizes

$$\begin{aligned}\begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} &= \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\end{aligned}$$

More seriously, even if A and B are square matrices of the same size, so that AB and BA are both well defined square matrices of this size, they may not be equal. An example coming from geometry takes A to be a quarter turn

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

and B to be a reflection

$$B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad BA = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

If $AB = BA$, we say that A and B *commute*. On the homework, I will ask you to work out some examples of matrices which do and do not commute.

A Algebraic Properties of Matrix Vector Algebra

Here I summarize the algebraic properties of vector addition and scalar multiplication, matrix-vector multiplication, and matrix multiplication.

Below, $a, b, c \in \mathbb{R}$ are scalars, u, v, w are vectors, and A, B, C are matrices. When I write a product, I implicitly assume that the matrices and vectors have sizes so that it is well defined.

$$a(bv) = (ab)v, \quad (a + b)v = av + bv, \quad a(v + w) = av + aw$$

$$A(v + w) = Av + Aw, \quad (A + B)v = Av + Bv$$

$$A(bv) = (bA)v = b(Av)$$

$$A(Bv) = (AB)v$$

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC$$

$$A(cB) = (cA)B = c(AB)$$

$$AI_n = I_m A = A \quad (\text{here } A \text{ is an } m \times n \text{ matrix})$$